Truncated Whitehead principle in terms of loop spaces: a bijection on components plus loop-space equivalences give an equivalence

The Univalent Foundations Program

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Editorial note. The counted statement is the author's 'thm:whitehead1' (source lines 2398-2405, Corollary 8.8.2); the proof environment starts on the next line, so the pairing is by adjacency.

Editorial note. The retained context is the author's Theorem 8.8.1, invoked by label at the first line of the retained proof; retaining it keeps that label defined inside the package.

Editorial note. The retained counter values (chapter 8, section 8, theorem offset 0) make the retained context print as Theorem 8.8.1 and the counted corollary as Corollary 8.8.2, matching the original book numbering.

Editorial note. The single external reference is the equivalence-inhabited-codomain result of section 4.4, whose number is taken from the label-number dump of the pinned book snapshot.

Editorial scope. One original result of the HoTT Book, counted once: the truncated Whitehead principle in terms of loop spaces (Corollary 8.8.2).

Allowed standard prerequisites (editorial).

- Loop space of a pointed type. Omega(A,x) is the pointed type of loops at x, and Omega f is the induced map of loop spaces.. It is the object of the second hypothesis and of the commuting square in the proof.
- Theorem 8.8.1 (truncated Whitehead, base case). If trunc 0 f is surjective and ap f is an equivalence on all path spaces then f is an equivalence.. It reduces the goal to showing that ap f is an equivalence, which is the first line of the proof.
- Two-out-of-three property for equivalences. If two of three maps in a commuting square of equivalences are equivalences then so is the third.. It is applied to the displayed square at the end of the proof.

Source reference. Original theorem 4.4.6: sources/hott-homotopy/book-labelnumbers-first-edition.tex, source lines 1774–1774. Original result cited by the retained text but located outside the retained span. It is a navigation pointer: the counted statement's own author proof is retained line for line, and no part of

it is replaced by this reference. The reference number is taken from the label-number dump of the pinned book snapshot.

Theorem 8.8.1. *Suppose $f : A \to B$ is a function such that*

(i) $\|f\|_0 : \|A\|_0 \to \|B\|_0$ *is surjective, and*
(ii) for any $x, y : A$, the function $\mathsf{ap}_f : (x =_A y) \to (f(x) =_B f(y))$ is an equivalence.

Then f is an equivalence.

Corollary 8.8.2. *Suppose $f : A \to B$ is a function such that*

(i) $\|f\|_0 : \|A\|_0 \to \|B\|_0$ *is a bijection, and*
(ii) for any $x : A$, the function $\Omega f : \Omega(A, x) \to \Omega(B, f(x))$ is an equivalence.

Then f is an equivalence.

Proof. By Theorem 8.8.1, it suffices to show that $\mathsf{ap}_f : (x =_A y) \to (f(x) =_B f(y))$ is an equivalence for any $x, y : A$. And by original theorem 4.4.6, we may assume $f(x) =_B f(y)$. In particular, $|f(x)|_0 = |f(y)|_0$, so since $\|f\|_0$ is an equivalence, we have $|x|_0 = |y|_0$, hence $|x = y|_{-1}$. Since we are trying to prove a mere proposition (ap_f being an equivalence), we may assume given $p : x = y$. But now the following square commutes up to homotopy:

$$\begin{array}{ccc} \Omega(A, x) & \xrightarrow{- \cdot p} & (x =_A y) \\ {\scriptstyle \Omega f}\downarrow & & \downarrow{\scriptstyle \mathsf{ap}_f} \\ \Omega(B, f(x)) & \xrightarrow[- \cdot f(p)]{} & (f(x) =_B f(y)). \end{array}$$

The top and bottom maps are equivalences, and the left-hand map is so by assumption. Hence, by the 2-out-of-3 property, so is the right-hand map. □